Compiler Construction

Lab-6

Syeda Ayesha Fatima - 53953

BSCS-vi

Submitted to: Ma’am Hasnat

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Task 1:

```
Enter number of productions: 5
Enter productions (Example: E=TR):
S=AB
A=aA
A=$
B=bB
B=c

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b c }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b c }
FOLLOW(B) = { $ }
Continue? (y/n):
```

Task 2:

```
E=TR
R=+TR
R=$
T=FY
Y=*FY
Y=$
F=(E)
F=ID

Find FIRST and FOLLOW of: F
FIRST(F) = { ( }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: Y
FIRST(Y) = { * $ }
FOLLOW(Y) = { + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: T
FIRST(T) = { ( }
FOLLOW(T) = { + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: R
FIRST(R) = { + $ }
FOLLOW(R) = { $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: E
FIRST(E) = { ( }
FOLLOW(E) = { $ ) }
```

Task 3:

```
Enter number of productions: 6
Enter productions (Example: E=TR):
S=ABC
A=a
A=$
B=b
B=$
C=c

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b c }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c }
Continue? (y/n): y

Find FIRST and FOLLOW of: C
FIRST(C) = { c }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Task 4:

```
Enter number of productions: 4
Enter productions (Example: E=TR):
S=aS
S=bA
A=cA
A=d

Find FIRST and FOLLOW of: A
FIRST(A) = { c d }
FOLLOW(A) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Task 5:

```
Enter number of productions: 2
Enter productions (Example: E=TR):
S=(S)S
S=$

Find FIRST and FOLLOW of: S
FIRST(S) = { ( $ }
FOLLOW(S) = { $ ) }
Continue? (y/n):
```

Task 6:

```
Enter number of productions: 6
Enter productions (Example: E=TR):
S=AB
A=a
A=$
B=bC
C=c
C=$

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b }
FOLLOW(B) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: C
FIRST(C) = { c $ }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Task 7:

```
Enter number of productions: 8
Enter productions (Example: E=TR):
E=TR
R=+TR
R=$
T=FS
S=*FS
S=$
F=i
F=(E)

Find FIRST and FOLLOW of: E
FIRST(E) = { i ( }
FOLLOW(E) = { $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: R
FIRST(R) = { + $ }
FOLLOW(R) = { $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: T
FIRST(T) = { i ( }
FOLLOW(T) = { + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { * $ }
FOLLOW(S) = { + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: F
FIRST(F) = { i ( }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): n
```

Task 8:

```
Enter number of productions: 5
Enter productions (Example: E=TR):
S=AB
A=Ba
A=$
B=b
B=$

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { $ a }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { b a $ }
FOLLOW(A) = { b $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { b a $ }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Task 9:

```
Enter number of productions: 7
Enter productions (Example: E=TR):
S=ABC
A=aA
A=$
B=bB
B=$
C=cC
C=d

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b c d }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c d }
Continue? (y/n): y

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c d }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Task 10:

```
Enter number of productions: 7
Enter productions (Example: E=TR):
S=ACB
A=aA
A=$
B=bB
B=$
C=c
C=d

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { c d }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { b $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a c d }
FOLLOW(S) = { $ }
Continue? (y/n):
```

Compiler Construction (LAB)

Name: Syeda Ayesha Fatima

SAP: 53953

1. $S \rightarrow AB$

$A \rightarrow aA / \varepsilon$

$B \rightarrow bB / c$

First: First(A) = $\{a, \varepsilon\}$

First(B) = $\{b, c\}$

First(S) = $\{a, b, c\}$

Follow: Follow(S) = $\{\$\}$

Follow(A) = $\{b, c\}$

Follo(B) = $\{\$\}$ = Follow(S)

2. $E \rightarrow TR$

$R \rightarrow +TR / \varepsilon$

$T \rightarrow FY$

$Y \rightarrow *FY / \varepsilon$

$F \rightarrow (E) / id$

First:

first(F) $\Rightarrow$ $\{(, id\}$ First(E) = $\{(, id\}$

First(Y) = $\{*, \varepsilon\}$

First(T) = $\{(, id\}$

Firs(R) = $\{+, \varepsilon\}$

6.

First(A) = {a, ε}	Follow(S) = {$}
First(B) = {b}	Follow(A) = {b}
First(C) = {e, ε}	Follow B = {c}
~~First(S) = {a, b}~~	Follow(C) = {$}

First(S) = {a, b}

7.

First(F) = {id, (}	Follow(E) = {), $}
First(T') = {*, ε}	Follow(E') = {), $}
First(T) = {id, (}	Follow(T) = {+,), $}
First(E') = {+, ε}	Follow(T') = {+,), $}
First(E) = {id, (}	Follow(F) = {*, +,), $}

8.

First(A) = {a, ε}	Follow(S) = {$}
First(B) = {b, ε}	Follow(A) = {b, ε}
	Follow(B) = {$}
First(S) = {a, b}	

9.

First(A) = {a, ε}	Follow(S) = {$}
First(B) = {b, ε}	Follow(A) = {b, ε}
First(C) = {c, d}	Follow(B) = {c, d}
First(S) = {a, b, c, d}	Follow(C) = {$}

6. $S \rightarrow ACB$

First (A) = {a, ε}
First (B) = {b, ε}
First (C) = {c, d}

First (S) = {a, c, d}

Follow (A) = {c, d}
Follow (S) = {$}
Follow (B) = {$}